



ME 203: Energy Resources & Conversion

Course Code	ME 203
Course Name	Energy Resources & Conversion
L-T-P-C	3-0-0-3
Prerequisites	Nil
Approval	5th Senate (Oct 2013)

Course Contents

- **Introduction:** Energy resources spectrum, Renewable and non Renewable energy Sources, consumption pattern in various sectors, Efficiency of energy resources, load demand, and economics (4 hrs)
- **Coal:** Classification, properties, combustion, carbonization, liquefaction and gasification, Electricity generation from coal(4 hrs)
- **Liquid fuels:** various type of fuels, properties and handling (3 hrs)
- **Gaseous fuel:** CNG, LNG, and LPG (3 hrs)
- **Nuclear Energy:** Potential, Fusion and fission processes and nuclear reactor (4 hrs)
- **Wind:** potential and utilization (2 hrs)
- **Solar Energy:** Solar radiation measurements, Solar Thermal: Flat plate and focusing collectors, solar space heating and cooling, solar pond, Solar Photovoltaic: Solar cells and storage (5 hrs)
- **Hydropower:** classification, components of hydropower generation systems (3 hrs)
- **Biomass Energy:** Biomass Types, characterization, conversion routs, bio chemical (4 hrs)
- **Other Energy Resources:** Hydrogen, Fuel Cells and other energy sources (4 hrs)
- **Energy Conservation:** Waste heat recovery, use of low grade hot streams, concept of vapour recompression, flash vaporization, heat pipe, energy targeting by pinch method (6 hrs)

Text and Reference Books

1. Rao S and Parulekar BB, **Energy Technology: non conventional, renewable and conventional**, Khanna Publishers, 1995.
2. Rai, GD, **Non-conventional energy sources**, Khanna Publishers, 1994.
3. **World Energy Outlook 2006**, International Energy Agency, France, 2006.
4. **Electricity in India**, International Energy Agency, France, 2002.
5. Twidel, J. and Tonyweir, **Renewable Energy Resources**, Second Edition, Taylor & Francis, 2006.
6. Manwell, JF et.al., Wiley, **Wind Energy Explained – Theory, Design & Application**, 2002.
7. Sukhatme, S P, **Solar Energy: Principles of Thermal Collection and Storage**, Second Edition, Tata McGraw-Hill Publishing Company Ltd, 2008.
8. Takahashi, Peytrick & Trenka, Andrew; Wiley **Ocean Thermal Energy Conservation**, 1996.
9. Teddy, **DIRECTORY**, TERI, New Delhi, 2006.